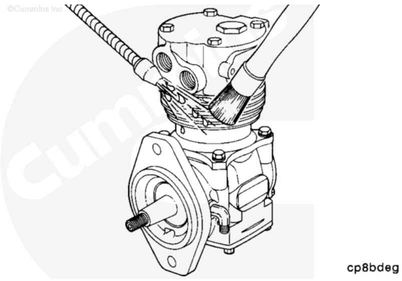
Trainings

Disassemble



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

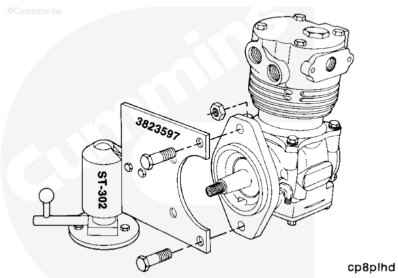
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to clean the air compressor.

Dry the air compressor with compressed air.

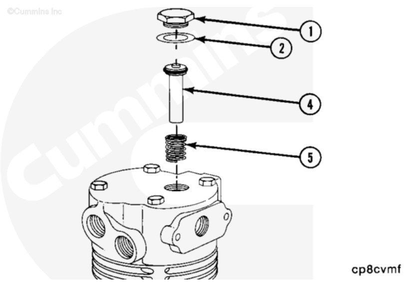
Mount the compressor to the mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



Disassemble the air compressor cylinder head and remove the following parts. Refer to Procedure 012-101 (/qs3/pubsys2/xml/en/procedures/04/04-012-101.html).

- Unloader valve (1).
- Copper washer (2).
- Unloader pin (4).
- Spring (5).

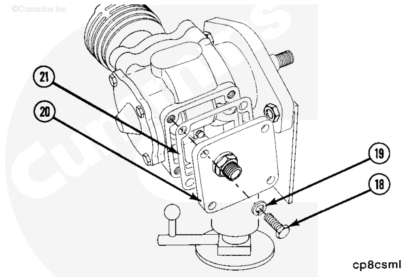
Discard the copper washer.



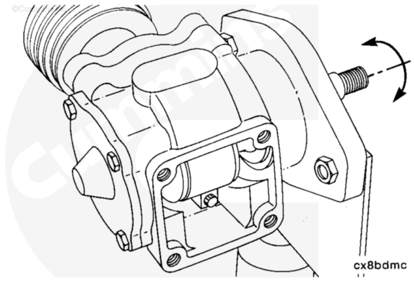
Remove the four bottom plate capscrews (18) and washers (19).

Remove the bottom cover plate (20).

Remove and discard the gasket (21).



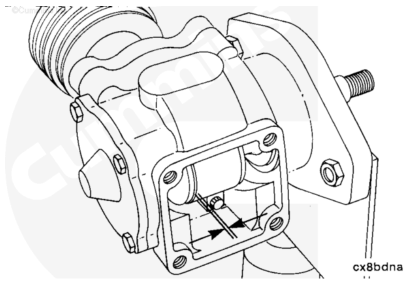
Rotate the crankshaft so the connecting rod journal is at bottom dead center (BDC).



Measure the side clearance between the connecting rod and the crankshaft.

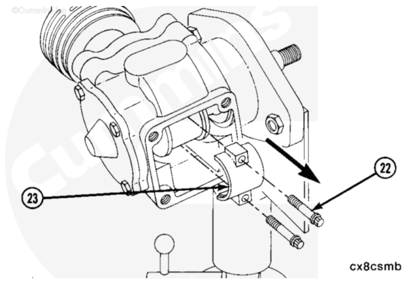
Replace the connecting rod if the clearance between it and the crankshaft exceeds 0.2540 mm [0.010 in].

Note : If the connecting rod needs to be replaced, the piston and connecting rod must be replaced as an assembly.

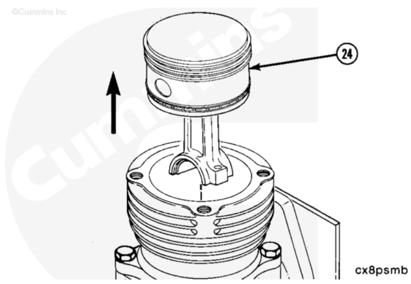


Remove the two 12-point connecting rod capscrews (22).

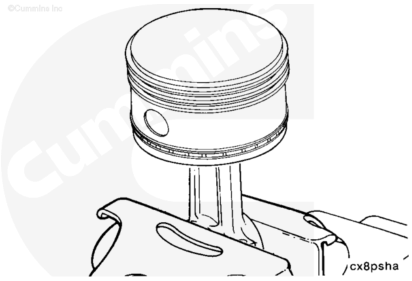
Remove the connecting rod cap (23).



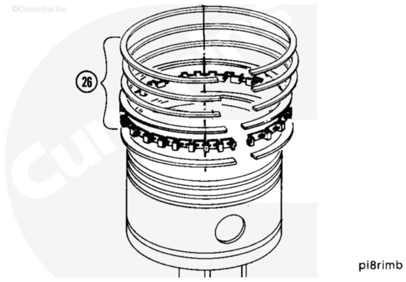
Remove the piston (24) and connecting rod from the crankcase.



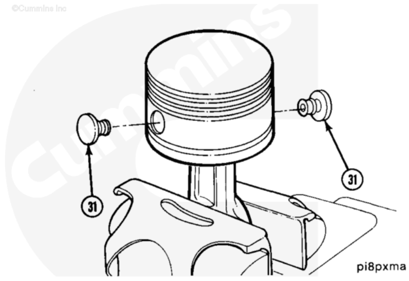
Install the piston and connecting rod assembly into a soft jawed vise.



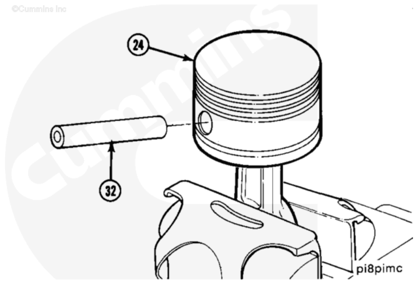
Remove the piston rings (26).



Remove the two buttons (31) from the piston wrist pin holes.



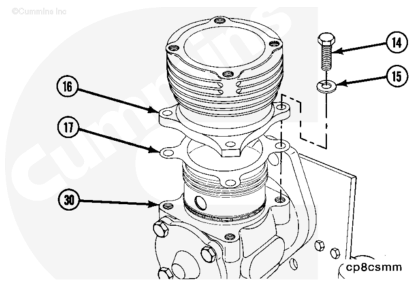
Remove the wrist pin (32) from the piston (24).



Remove the four cylinder block capscrews (14) and washers (15).

Remove the cylinder block (16) from the crankcase (30).

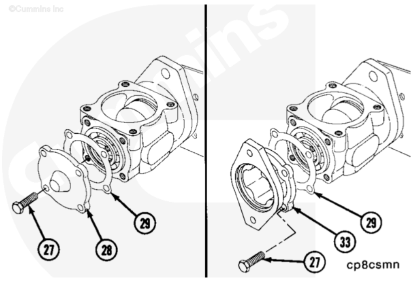
Remove and discard the gasket (17).



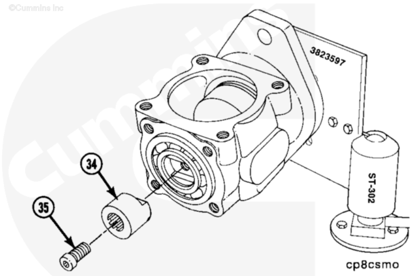
Remove the four capscrews (27) from the bearing cap (28) or power steering adapter (33).

Remove the bearing cap or power steering adapter.

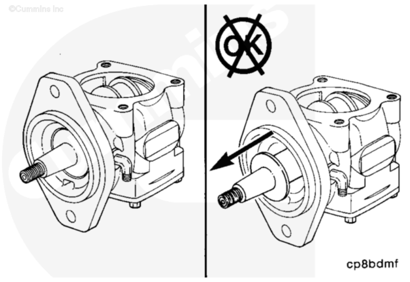
Remove and discard the gasket (29).



Remove the capscrew (35) and the power steering coupling (34).



The crankshaft and bearings can **not** be removed from the crankcase. If these parts are damaged, they **must** be replaced as an assembly.



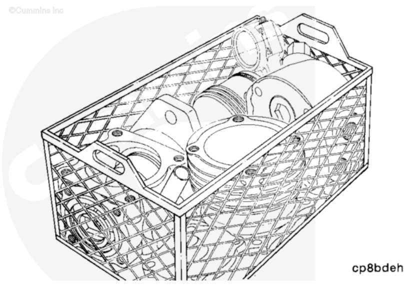
Clean and Inspect for Reuse

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

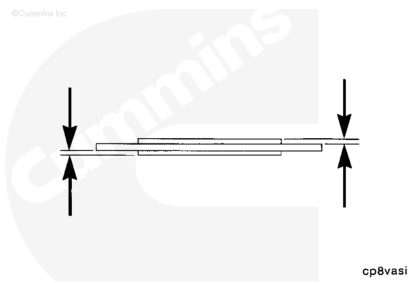
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Soak the parts in a kerosene emulsion based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

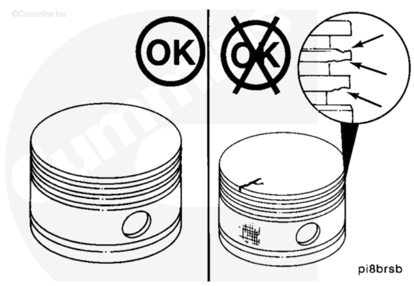
Inspect the air compressor cylinder head components. Refer to Procedure 012-101 (/qs3/pubsys2/xml/en/procedures/04/04-012-101.html).



Inspect the piston top and pin bore for cracks.

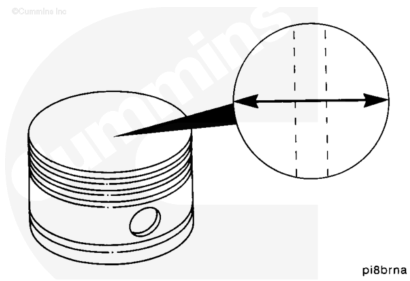
Check the ring grooves and skirt for damage. Replace, if damaged.

Note : If the piston needs to be replaced, the piston and connecting rod must be replaced as an assembly.



Measure the outside diameter of the piston at 90 degrees from the piston pin bore and below the compression rings.

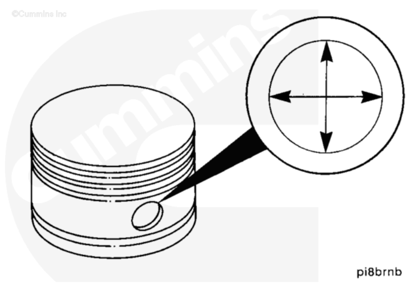
mm		in
79.858	MIN	3.1440
79.883	MAX	3.1450



Replace the piston and connecting rod assembly, if **not** within specification.

Measure the piston pin bore.

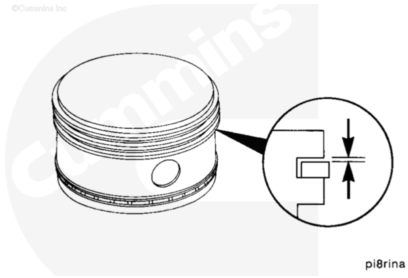
mm		in
14.290	MIN	0.5626
14.295	MAX	0.5628



Use new piston rings and a feeler gauge to measure for ring groove wear.

Replace the piston and connecting rod assembly if the gaps are worn larger than the dimensions listed below:

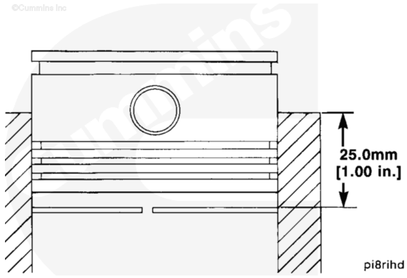
	mm		in
Compression Ring	0.010	MIN	0.0004
	0.058	MAX	0.0023
Oil Ring	0.013	MIN	0.0005
	0.114	MAX	0.0045



Note : Cummins Inc. recommends that new piston rings be installed during rebuild.

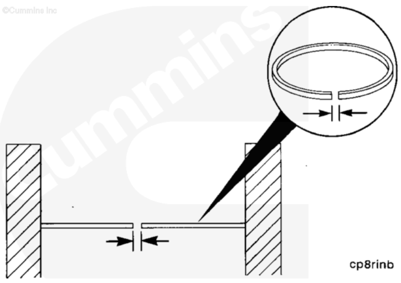
If old rings are used, follow the instructions listed below to measure the ring gaps.

Insert on ring at a time into the cylinder bore. Seat the ring with a piston head squarely 25.00 mm [1.0 in] below the top of the crankcase.



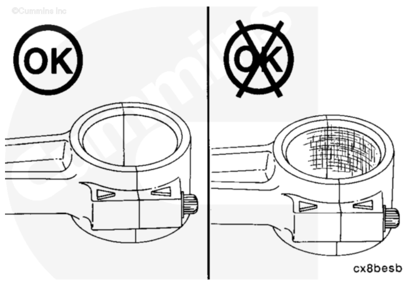
If the ring gap exceeds the specifications listed below, the rings **must** be replaced.

	mm		in
Compression Ring	0.020	MIN	0.0008
	0.051	MAX	0.0020
Oil Ring	0.035	MIN	0.0015
	1.40	MAX	0.0055

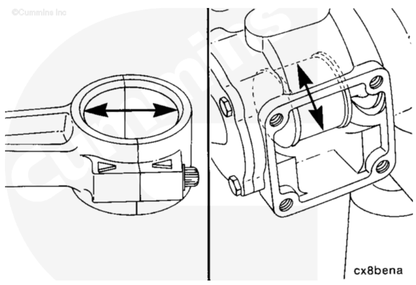


Note : Cummins Inc. recommends that new bearings be installed during rebuild.

Inspect the connecting rod bearing for scoring or wear.



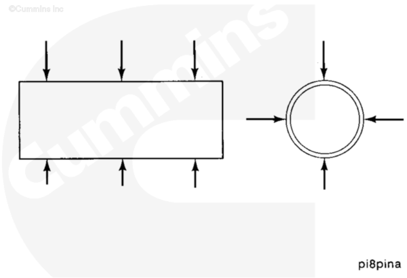
Replace the bearings if the clearance between the bearings and crankshaft journal exceeds 0.1270 to 0.5334 mm [0.005 too 0.0021 in].



Measure the piston pin outside diameter.

mm		in
14.270	MIN	0.5618
14.275	MAX	0.5620

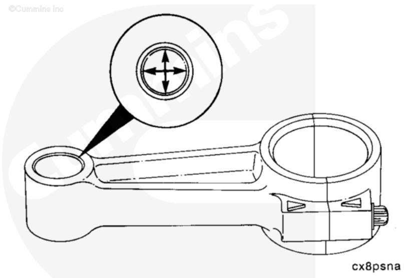
Replace the piston and connecting rod assembly, if **not** within specification.



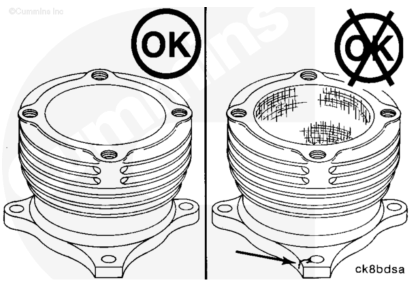
Measure the inside diameter of the piston end of the connecting rod.

mm		in
14.290	MIN	0.5625
14.303	MAX	0.5631

Replace the piston and connecting rod, if **not** within specification.



Inspect and replace the cylinder block if cracks, scoring, or other damage is found.



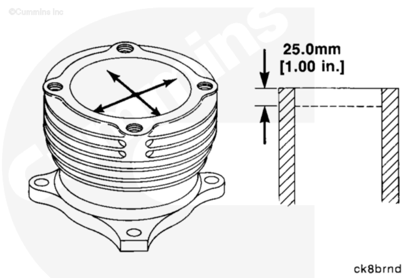
Use a dial bore gauge, Part Number 3375072, or equivalent, to measure the cylinder bore.

Measure at 25.0 mm [1.00 in] below the top of the block.

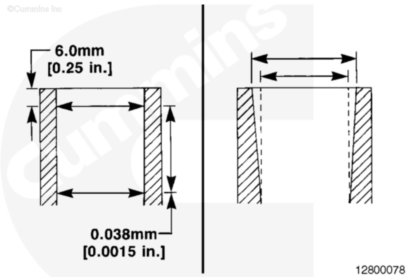
Replace the block, if **not** with specification.

Maximum out-of-round is 0.013 mm [0.0005 in].

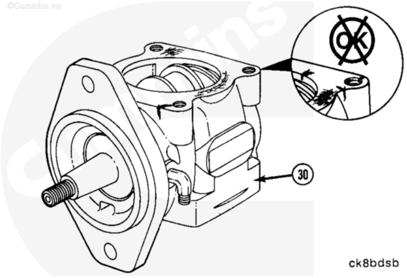
mm		in
80.013	MIN	3.1501
80.038	MAX	3.1511



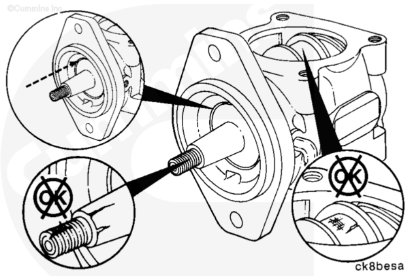
Maximum taper in the piston travel area is 0.038 mm [0.0015 in].



Inspect and replace the crankcase assembly, if any damage is found.

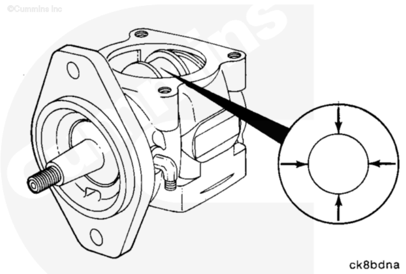


Inspect and replace the crankcase assembly if the crankshaft or roller bearings are damaged.



Measure the crankshaft journal diameter.

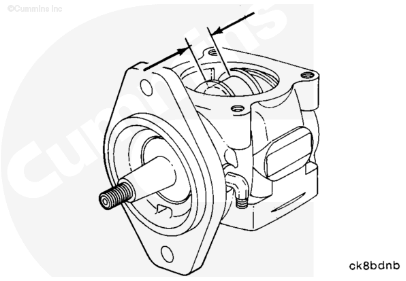
mm		in
30.099	MIN	1.1850
30.112	MAX	1.1855



Replace the crankcase assembly, if **not** within specification.

Measure the crankshaft journal width.

mm		in
32.13	MIN	1.265
32.18	MAX	1.267

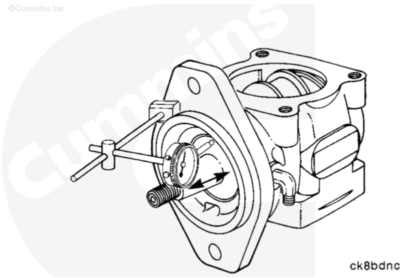


Replace the crankcase assembly, if **not** within specification.

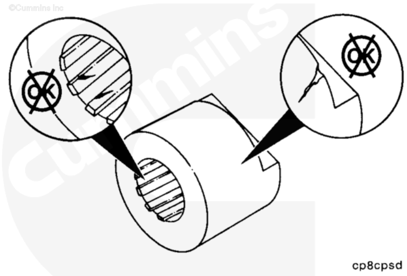
Measure the crankshaft end play.

mm		in
0.03	MIN	0.001
0.81	MAX	0.032

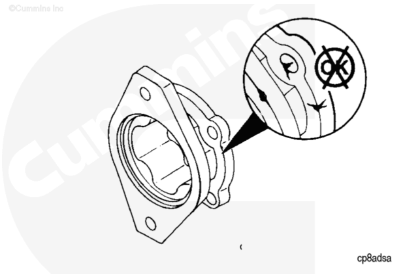
Replace the crankcase assembly, if **not** within specification.



Inspect the power steering coupling for wear or cracks. Replace the coupling, if damaged.

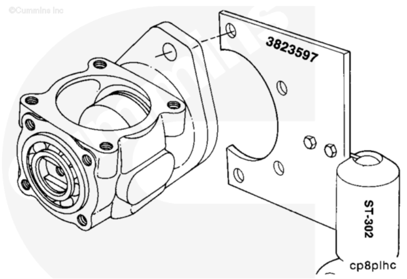


Inspect and replace the power steering adapter, if any damage is found.



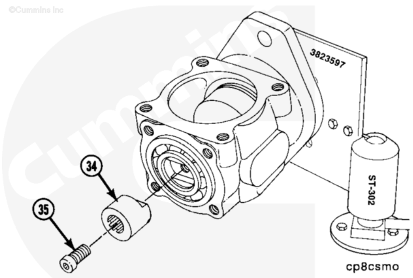
Assemble

Install the crankcase to the mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



Install the power steering (34) coupling and capscrew (35) into the crankshaft.

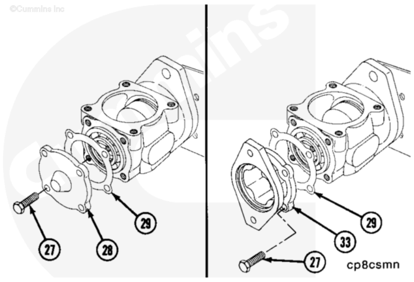
Torque Value: 23 n•m [17 ft-lb]



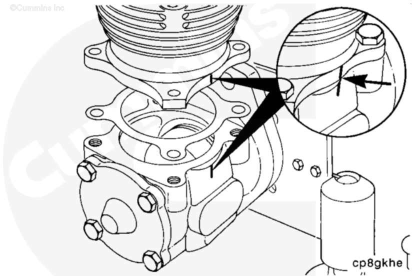
Install a new gasket (29) and the bearing cap (28) or power steering adapter (33). Align the scribe markds with the mark on the crankcase.

Install the four capscrews (27) and tighten the capscrews.

Torque Value: 16 n•m [12 ft-lb]

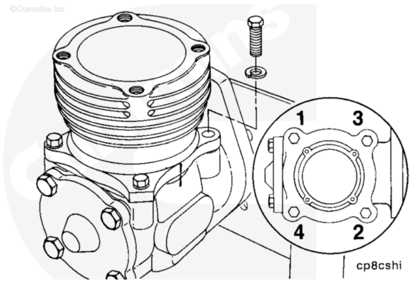


Install a new gasket and the cylinder block to the crankcase with the scribed marks aligned.

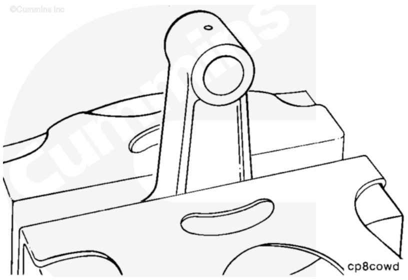


Install the four capscrews and tighten in the following sequence:

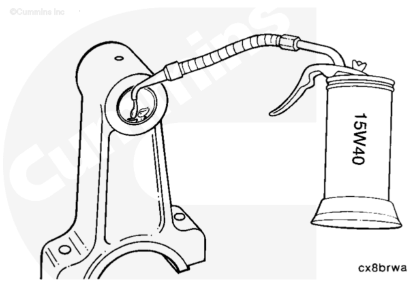
Torque Value:	Step 1	20 n.m [15 ft-lb]
	Step 2	41 n.m [30 ft-lb]



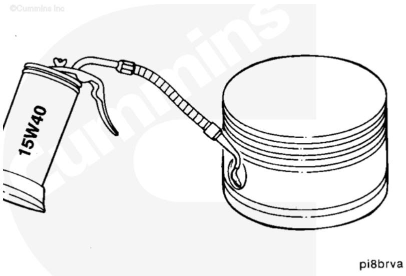
Install the connecting rod into a soft jawed vise.



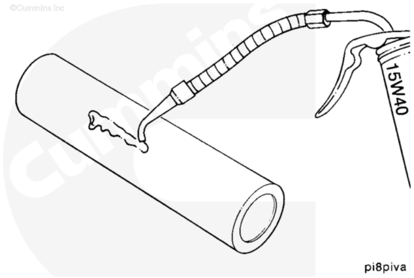
Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore of the connecting rod.



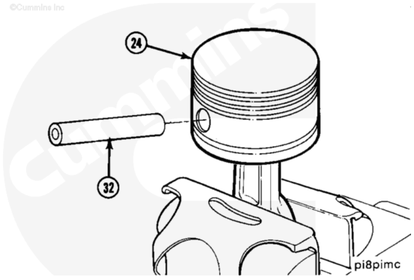
Use clean 15W-40 lubricating engine oil to lubricate the piston pin bore.



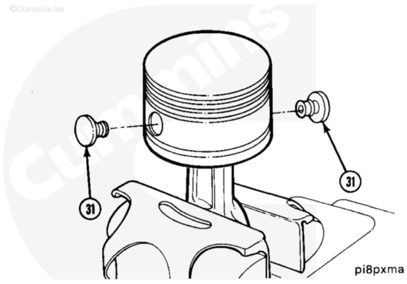
Use clean 15W-40 lubricating engine oil to lubricate the piston pin outside diameter.



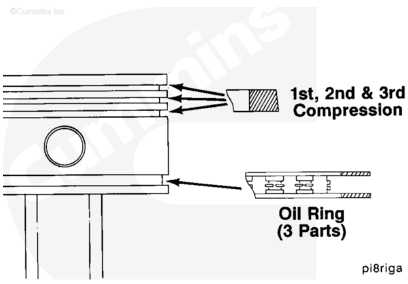
Install the piston pin.



Install the two buttons.



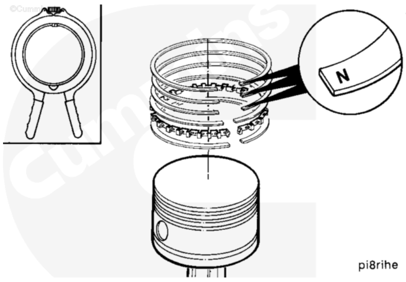
Piston ring positions.



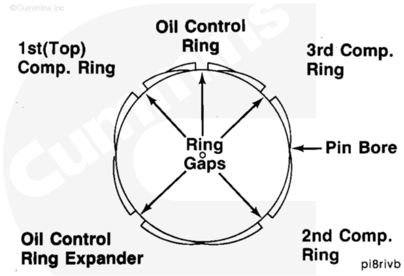
Install the compression rings with the top up. Top is identified with the letter “N” or a dot near the end of the ring gap.

Install the piston oil ring. If installing a three piece ring, first install the center expander and then the two rails.

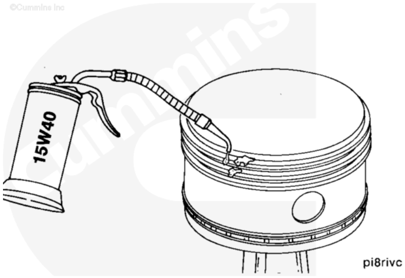
Note : The oil ring has no top or bottom.



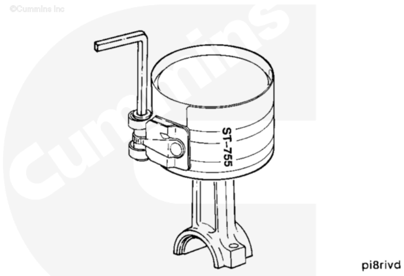
Position the ring gaps apart and make sure they are **not** over the piston pin bores.



Use clean 15W-40 lubricating engine oil to lubricate the rings.

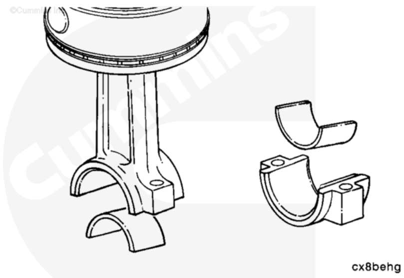


Use a piston ring compressor, Part Number ST-755, or equivalent, to compress the rings.



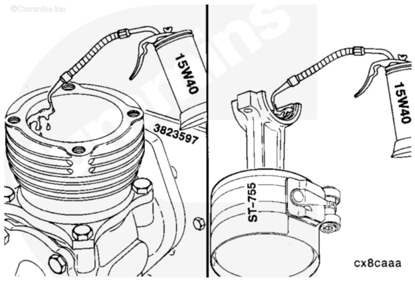
Note : Cummins Inc. recommends that new bearings be installed during rebuild.

Install the connecting rod bearings into the connecting rod and rod cap.

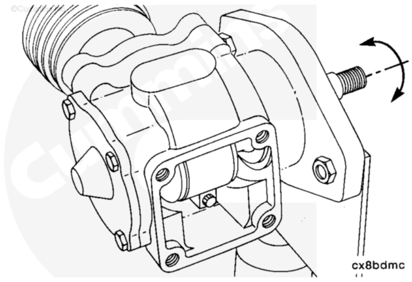


Use clean 15W-40 lubricating engine oil to lubricate the connecting rod and rod cap.

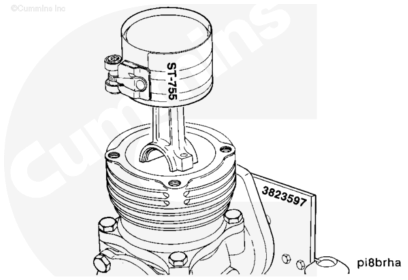
Use clean 15W-40 lubricating engine oil to lubricate the cylinder block bore.



Rotate the crankshaft so the connecting rod journal is at bottom dead center (BDC).



Carefully install the piston and connecting rod assembly into the cylinder block bore.

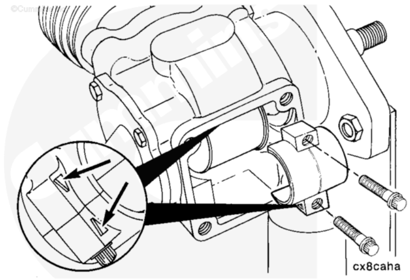


Install the connecting rod cap aligning the arrows.

Install the two 12-point capscrews into the cap and connecting rod.

Tighten the capscrews.

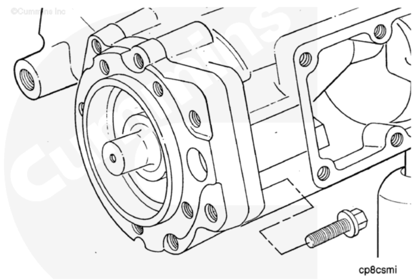
Torque Value: 25 n•m [18 ft-lb]



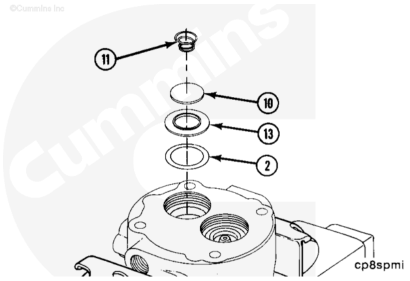
Install a new gasket (21) and the bottom cover plate (20).

Install the four washers (19) and capscrews (18) and tighten the capscrews.

Torque Value: 16 n•m [12 ft-lb]



Assemble the air compressor cylinder head. Refer to Procedure 012-101 (/qs3/pubsys2/xml/en/procedures/04/04-012-101.html).



Last Modified: 13-Jun-2005